

Lab Project Report

Group 8

Project Overview: Cricket Player Position Prediction

[CSE422-Project/422 Project Test 2/notebooks/CSE422_Lab_Project_Report.ipynb at main · Tanjeem-Samir-Khan/CSE422-Project](#)

Group members:

Tanjeem Samir Khan - 22301058

Raiyan Ahmed - 22301639

Course: CSE422 - Artificial Intelligence

Section: 12

Table of contents:

1. Introduction	pg1
2. Dataset Description	pg1
3. Dataset Preprocessing	pg9
4. Dataset Splitting	pg10
5. Model Training & Testing	pg11
6. Model Selection & Comparison Analysis	pg13
7. Conclusion	pg17

1. Introduction:

Cricket is a team sport where players have different roles and responsibilities based on their position. Each position requires specific skills and characteristics. The primary objective of this project is to predict a cricket player's position based on their physical and playing characteristics using machine learning algorithms. We analyze two comprehensive datasets containing information about international and franchise cricket teams and players from 15+ countries across four continents.

2. Dataset Description:

Total Number of Features: 16

This is a **CLASSIFICATION** problem because:

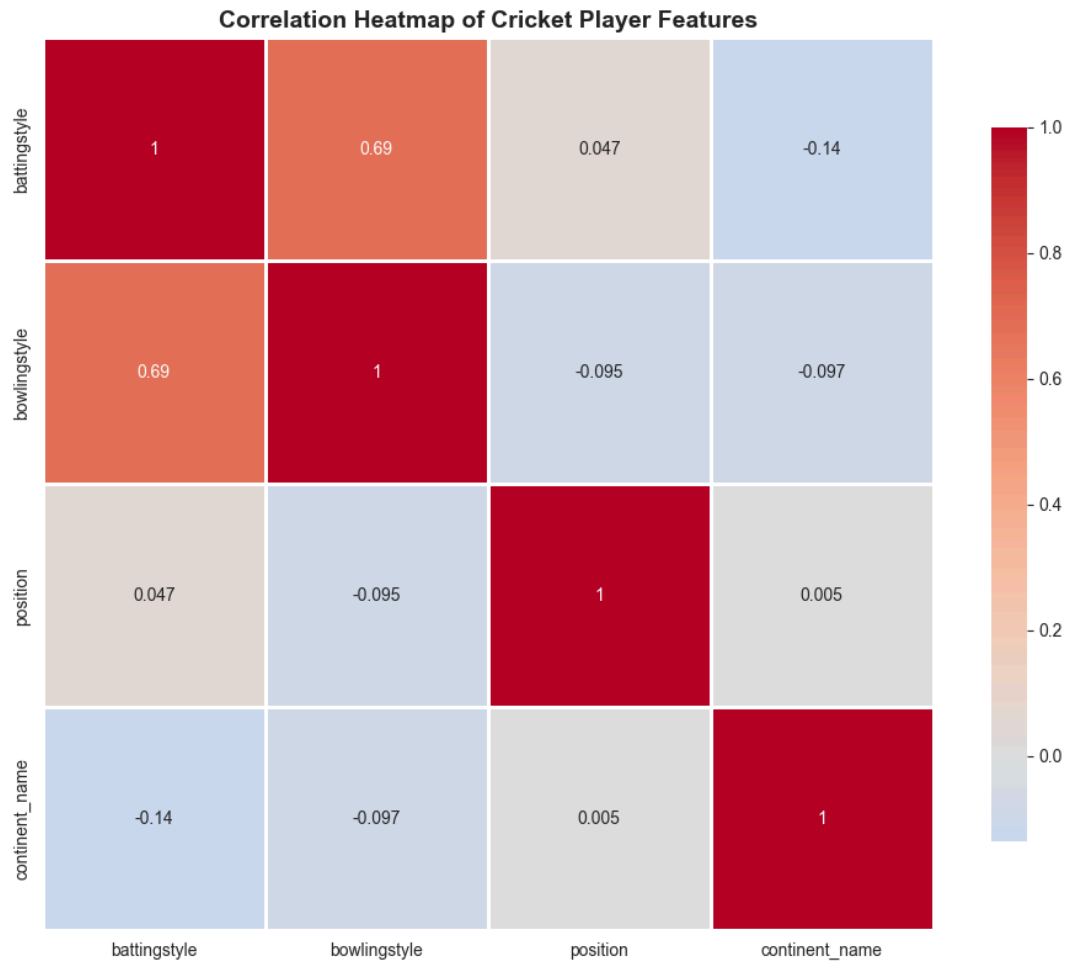
- The target variable (position) has discrete classes
- We aim to predict which category a player belongs to
- The output is categorical (Batsman, Bowler, Wicketkeeper, etc.), not continuous
- We use classification metrics like accuracy, precision, recall, and confusion matrix

Total Number of Data Points: 17385

Quantitative Features (3): ['id', 'continent_id', 'country_id']

Categorical Features (13): ['firstname', 'lastname', 'fullname', 'image_path', 'dateofbirth', 'gender', 'battingstyle', 'bowlingstyle', 'position', 'updated_at', 'continent_name', 'country_name', 'country_image_path']

Yes, for compatibility (Algorithm Requirement), Preventing False Ordering, Improving Model Accuracy and Enabling Semantic Interpretation



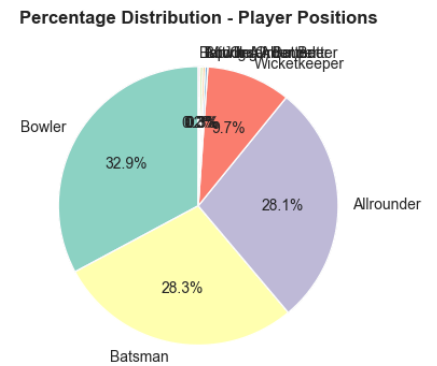
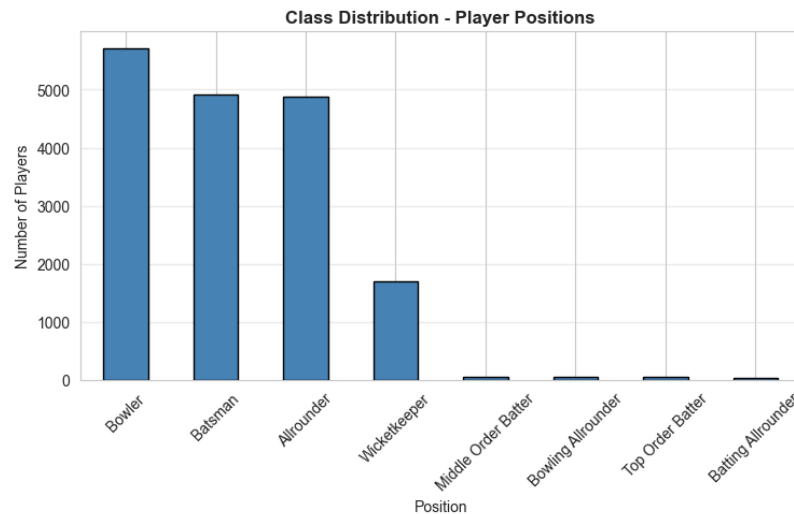
From the correlation heatmap, we observe:

- Different batsmen prefer different position
- Bowling Style shows strong correlation with position
- Continent may have indirect influence through regional playing styles

The correlation values guide us in feature selection and understanding which features are most predictive of player position.

Imbalance Dataset:

No, different classes have different number of instances



EXPLORATORY DATA ANALYSIS

1. BATTING STYLE DISTRIBUTION BY POSITION:

a.

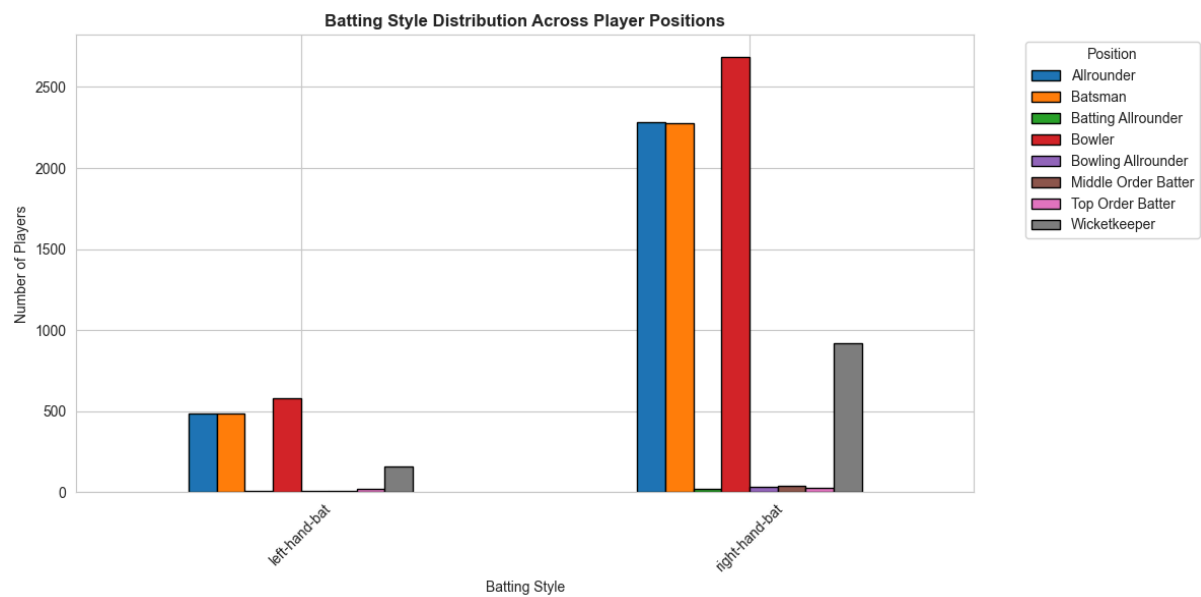
position	Allrounder	Batsman	Batting Allrounder	Bowler
battingstyle				
left-hand-bat	488	484	12	580
right-hand-bat	2285	2279	24	2686
All	2773	2763	36	3266

b.

position	Bowling Allrounder	Middle Order Batter	Top Order Batter
battingstyle			
left-hand-bat	12	11	19
right-hand-bat	32	43	30
All	44	54	49

c.

position	Wicketkeeper	All
battingstyle		
left-hand-bat	157	1763
right-hand-bat	921	8300
All	1078	10063



pg4

2. BOWLING STYLE DISTRIBUTION BY POSITION:

a.

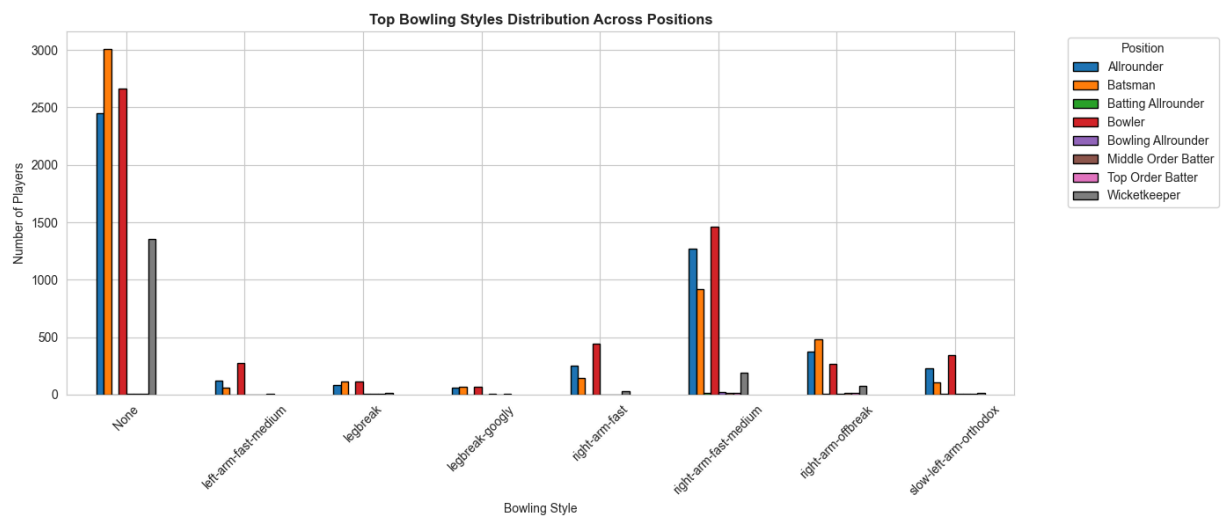
position	Allrounder	Batsman	Batting Allrounder	Bowler
bowlingstyle				
None	2446	3008	2	2667
left-arm-chinaman	1	0	0	1
left-arm-fast	29	6	0	71
left-arm-fast-medium	118	63	0	273
legbreak	82	115	2	117
legbreak-googly	60	65	0	65
right-arm-fast	253	147	0	440
right-arm-fast-medium	1274	918	16	1464
right-arm-offbreak	377	482	9	265
slow-left-arm-orthodox	228	106	9	344
slow-right-arm-orthodox	9	5	0	4
All	4877	4915	38	5711

pg5

b.

position	Bowling Allrounder	Middle Order Batter
bowlingstyle		
None	5	10
left-arm-chinaman	0	0
left-arm-fast	0	0
left-arm-fast-medium	2	1
legbreak	3	4
legbreak-googly	2	3
right-arm-fast	1	0
right-arm-fast-medium	23	16
right-arm-offbreak	8	16
slow-left-arm-orthodox	5	4
slow-right-arm-orthodox	0	0
All	49	54

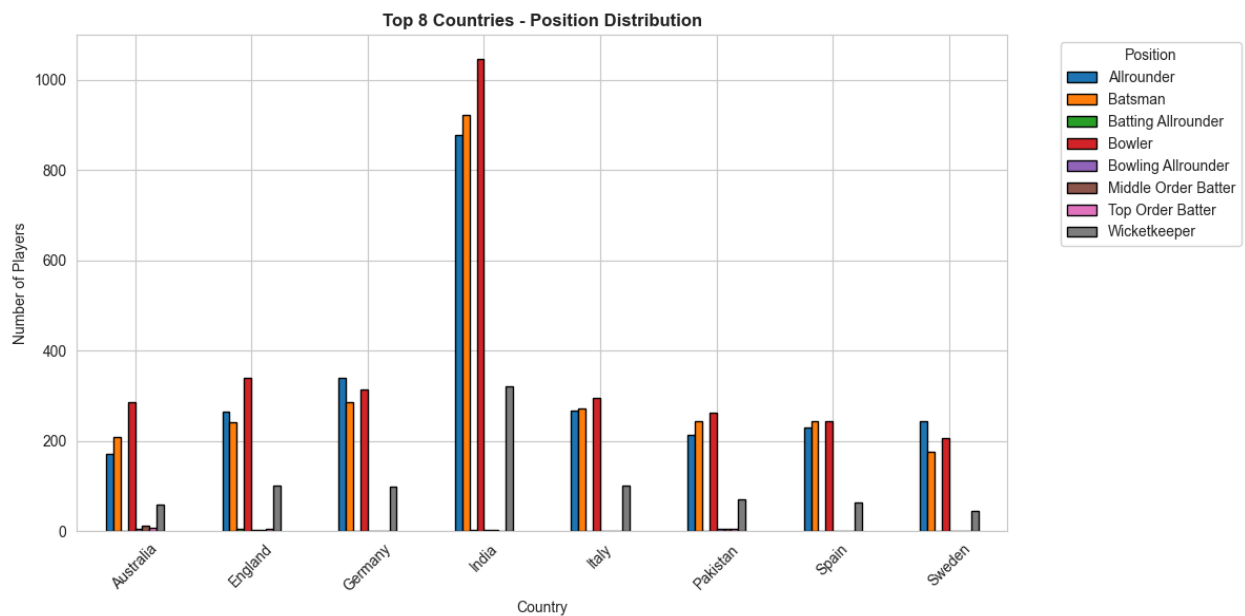
position	Top Order Batter	Wicketkeeper	All
bowlingstyle			
None	9	1355	9502
left-arm-chinaman	0	0	2
left-arm-fast	0	1	107
left-arm-fast-medium	0	8	465
legbreak	7	16	346
legbreak-googly	0	6	201
right-arm-fast	0	26	867
right-arm-fast-medium	11	191	3913
right-arm-offbreak	18	72	1247
slow-left-arm-orthodox	4	13	713
slow-right-arm-orthodox	0	4	22
All	49	1692	17385



pg7

3. TOP COUNTRIES BY NUMBER OF PLAYERS:

country_name	count
India	3179
Germany	1040
England	965
Italy	936
Pakistan	808
Spain	785
Australia	754
Sweden	672
Sri Lanka	481
Bangladesh	451



3. Dataset pre-processing

Faults:

a. Missing values:

Column	Missing_Count	Percentage
bowlingstyle	9502	54.656313
battingstyle	7322	42.116767

b. Categorical values:

Categorical features need conversion to numerical format for ML algorithms

c. Feature Scaling:

Improper feature scaling acts as a silent error that typically leads to overestimated performance during training but poor, unreliable results during actual deployment.

Solutions:

a. **'bowlingstyle'**: Filled missing values with **'No_Bowling_Style'** (players don't bowl)

b. **'battingstyle'** encoded

Mapping: {'left-hand-bat': np.int64(0), 'right-hand-bat': np.int64(1), nan: np.int64(2)}

'bowlingstyle' encoded

Mapping: {'left-arm-chinaman': np.int64(0), 'left-arm-fast': np.int64(1), 'left-arm-fast-medium': np.int64(2), 'legbreak': np.int64(3), 'legbreak-googly': np.int64(4), 'right-arm-fast': np.int64(5), 'right-arm-fast-medium': np.int64(6), 'right-arm-offbreak': np.int64(7), 'slow-left-arm-orthodox': np.int64(8), 'slow-right-arm-orthodox': np.int64(9), nan: np.int64(10)}

'gender' encoded

Mapping: {'f': np.int64(0), 'm': np.int64(1)}

'position' encoded

Classes: ['Allrounder' 'Batsman' 'Batting Allrounder' 'Bowler' 'Bowling Allrounder'

'Middle Order Batter' 'Top Order Batter' 'Wicketkeeper']

Encoding: {'Allrounder': np.int64(0), 'Batsman': np.int64(1), 'Batting Allrounder': np.int64(2), 'Bowler': np.int64(3), 'Bowling Allrounder': np.int64(4), 'Middle Order Batter': np.int64(5), 'Top Order Batter': np.int64(6), 'Wicketkeeper': np.int64(7)}

c. Selected Features for Modeling:

['battingstyle_encoded', 'bowlingstyle_encoded', 'gender_encoded']

Target Variable Shape: (17385,)

Applying StandardScaler:

Method: StandardScaler normalizes features to have mean=0 and std=1

Reason: Algorithms like KNN are distance-based and sensitive to feature scaling

Scaling Statistics:

Mean: [-7.84723643e-17 -1.56944729e-16 -2.12529320e-16]

Std Dev: [1.00002876 1.00002876 1.00002876]

Final Dataset Shape: (17385, 3)

Classes in Target Variable: 8

4. Dataset splitting

Strategy: Stratified (maintains class distribution in both sets)

Reason: Ensures balanced representation of all classes in both sets

Training Set Size: 13908 samples (80.0%)

Class Distribution in Training Set:

- Allrounder: 3902 samples (28.1%)
- Batsman: 3932 samples (28.3%)
- Batting Allrounder: 30 samples (0.2%)
- Bowler: 4569 samples (32.9%)
- Bowling Allrounder: 39 samples (0.3%)
- Middle Order Batter: 43 samples (0.3%)
- Top Order Batter: 39 samples (0.3%)
- Wicketkeeper: 1354 samples (9.7%)

Testing Set Size: 3477 samples (20.0%)

Class Distribution in Testing Set:

- Allrounder: 975 samples (28.0%)
- Batsman: 983 samples (28.3%)
- Batting Allrounder: 8 samples (0.2%)
- Bowler: 1142 samples (32.8%)
- Bowling Allrounder: 10 samples (0.3%)
- Middle Order Batter: 11 samples (0.3%)
- Top Order Batter: 10 samples (0.3%)
- Wicketkeeper: 338 samples (9.7%)

5. Model training & testing

a. K-Nearest Neighbors

Accuracy: 0.3181 (31.81%)

Precision: 0.3213

Recall: 0.3181

Confusion Matrix:

```
[[345 554    0  60    0    0    0  16]
 [259 683    0  33    0    0    0   8]
 [   3    5    0   0    0    0    0   0]
 [436 601    0  78    0    0    0  27]
 [   7    0    0   1    0    0    0   2]
 [   5    5    0   1    0    0    0   0]
 [   2    7    0   1    0    0    0   0]
 [  38 295    0   5    0    0    0   0]]
```

b. Decision Tree

Accuracy: 0.3733 (37.33%)

Precision: 0.3660

Recall: 0.3733

Confusion Matrix:

```
[[ 25 139    0  809    0    0    0    2]
 [ 29 249    0  698    0    0    0    7]
 [   0    2    0    6    0    0    0    0]
 [ 29 105    0 1007    0    0    0    1]
 [   3    1    0    6    0    0    0    0]
 [   1    4    0    6    0    0    0    0]
 [   3    5    0    2    0    0    0    0]
 [   2 161    0  158    0    0    0   17]]
```

Top 5 Important Features:

	Feature	Importance
1	bowlingstyle_encoded	0.715708
0	battingstyle_encoded	0.237709
2	gender_encoded	0.046583

c. Logistic Regression

Accuracy: 0.3428 (34.28%)

Precision: 0.2340

Recall: 0.3428

Confusion Matrix:

```
[[ 0 134  0 836  0  0  0  5]
 [ 0 179  0 783  0  0  0 21]
 [ 0  2  0  6  0  0  0  0]
 [ 0 141  0 993  0  0  0  8]
 [ 0  3  0  7  0  0  0  0]
 [ 0  3  0  8  0  0  0  0]
 [ 0  8  0  2  0  0  0  0]
 [ 0 150  0 168  0  0  0 20]]
```

d. Neural Network

Accuracy: 0.3753 (37.53%)

Precision: 0.4985

Recall: 0.3753

Confusion Matrix:

```
[[ 6 150  0 817  0  0  0  2]
 [ 2 260  0 714  0  0  0  7]
 [ 0  2  0  6  0  0  0  0]
 [ 0 119  0 1022  0  0  0  1]
 [ 0  3  0  7  0  0  0  0]
 [ 0  5  0  6  0  0  0  0]
 [ 0  6  0  4  0  0  0  0]
 [ 0 159  0 162  0  0  0 17]]
```

e. K-MEANS CLUSTERING

Number of Clusters (K): 8

Silhouette Score: 0.8440 (Range: -1 to 1, higher is better)

Adjusted Rand Index: 0.0186

Cluster Sizes:

Cluster 0: 6999 samples (40.3%)

Cluster 1: 4998 samples (28.7%)

Cluster 2: 418 samples (2.4%)

Cluster 3: 427 samples (2.5%)

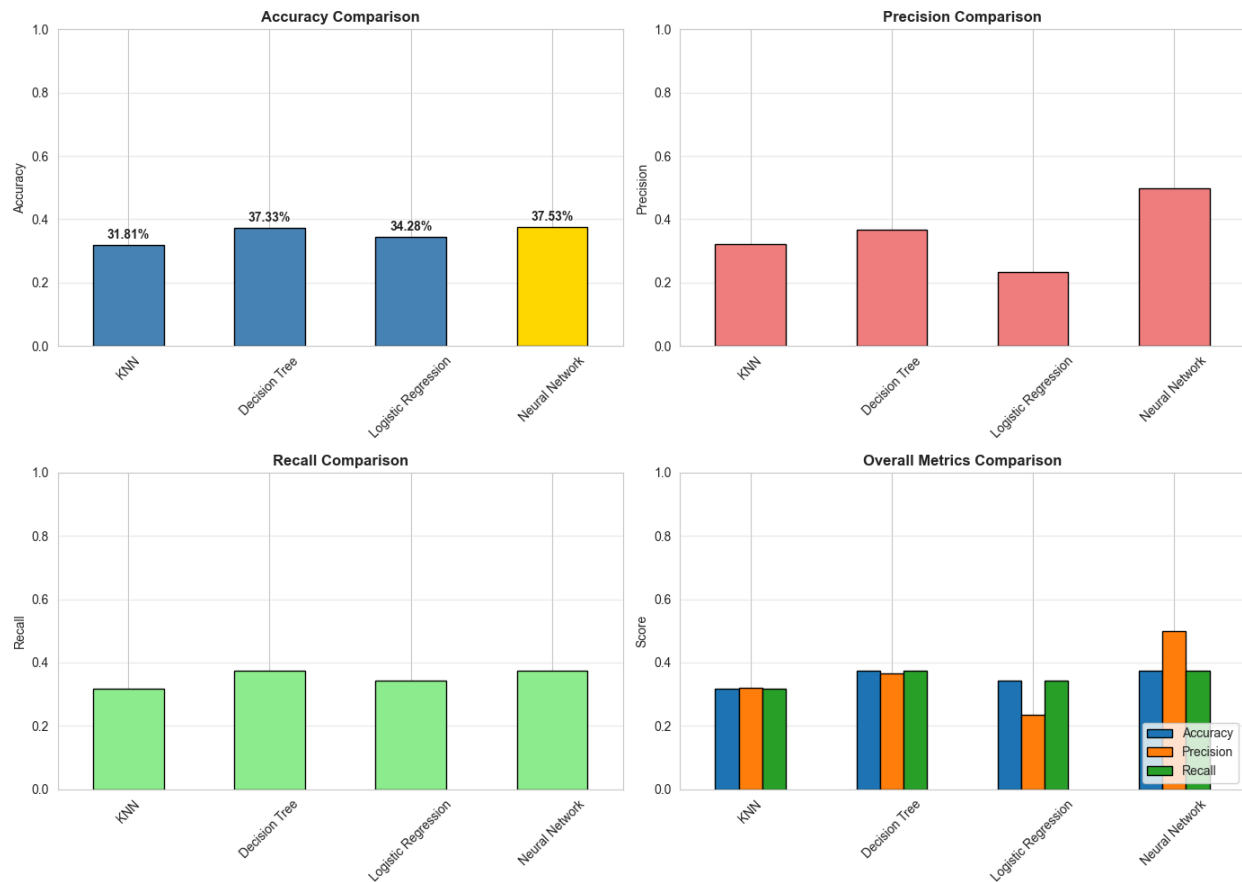
Cluster 4: 2043 samples (11.8%)

Cluster 5: 687 samples (4.0%)

Cluster 6: 1230 samples (7.1%)

Cluster 7: 583 samples (3.4%)

6. Model selection/Comparison analysis

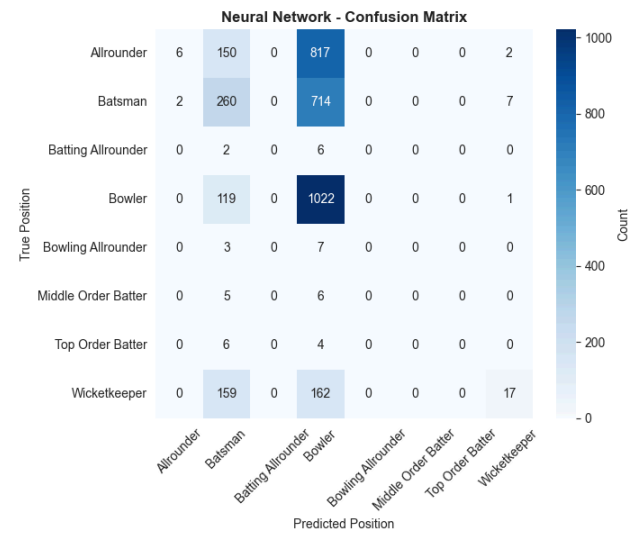
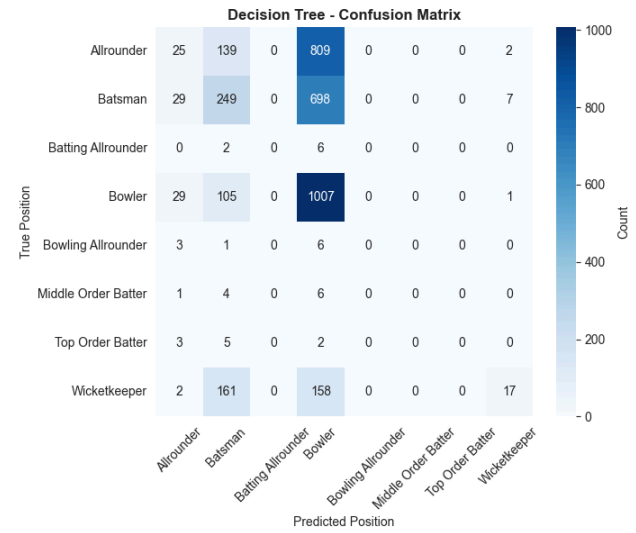
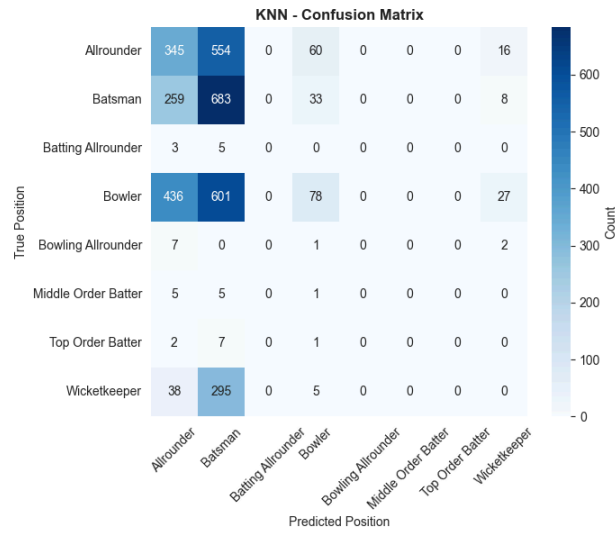


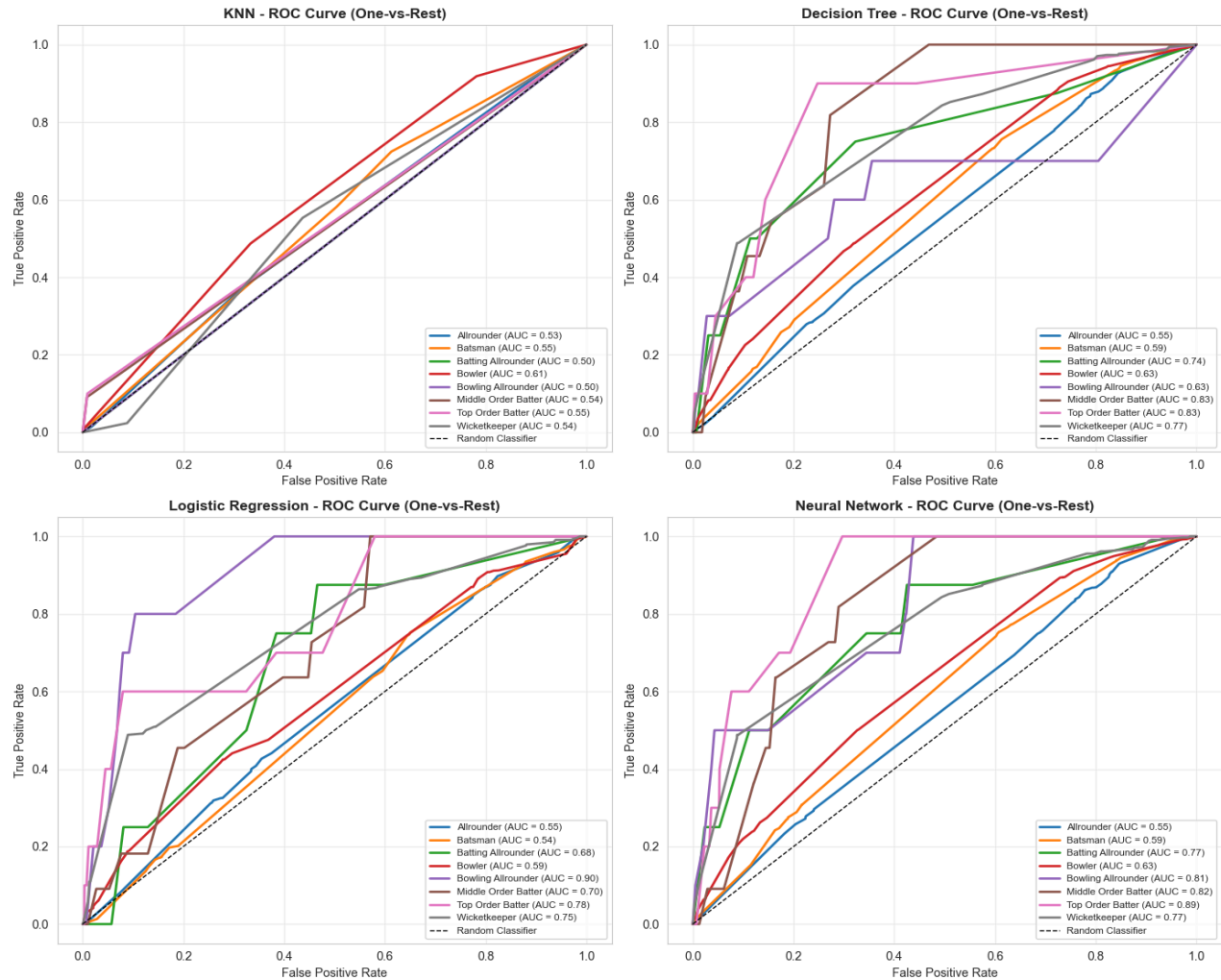
	Model	Accuracy	Precision	Recall
	KNN	0.318090	0.321282	0.318090
	Decision Tree	0.373310	0.365967	0.373310
	Logistic Regression	0.342824	0.233982	0.342824
	Neural Network	0.375324	0.498526	0.375324

RANKING BY ACCURACY:

1. Neural Network: 37.53%
2. Decision Tree: 37.33%
3. Logistic Regression: 34.28%
4. KNN: 31.81%

BEST MODEL: Neural Network
Accuracy: 37.53%





KNN - CLASSIFICATION REPORT

	precision	recall	f1-score	support
Allrounder	0.32	0.35	0.33	975
Batsman	0.32	0.69	0.44	983
Batting Allrounder	0.00	0.00	0.00	8
Bowler	0.44	0.07	0.12	1142
Bowling Allrounder	0.00	0.00	0.00	10
Middle Order Batter	0.00	0.00	0.00	11
Top Order Batter	0.00	0.00	0.00	10
Wicketkeeper	0.00	0.00	0.00	338
accuracy			0.32	3477
macro avg	0.13	0.14	0.11	3477
weighted avg	0.32	0.32	0.26	3477

DECISION TREE - CLASSIFICATION REPORT

	precision	recall	f1-score	support
Allrounder	0.27	0.03	0.05	975
Batsman	0.37	0.25	0.30	983
Batting Allrounder	0.00	0.00	0.00	8
Bowler	0.37	0.88	0.53	1142
Bowling Allrounder	0.00	0.00	0.00	10
Middle Order Batter	0.00	0.00	0.00	11
Top Order Batter	0.00	0.00	0.00	10
Wicketkeeper	0.63	0.05	0.09	338
accuracy			0.37	3477
macro avg	0.21	0.15	0.12	3477
weighted avg	0.37	0.37	0.28	3477

LOGISTIC REGRESSION - CLASSIFICATION REPORT

	precision	recall	f1-score	support
Allrounder	0.00	0.00	0.00	975
Batsman	0.29	0.18	0.22	983
Batting Allrounder	0.00	0.00	0.00	8
Bowler	0.35	0.87	0.50	1142
Bowling Allrounder	0.00	0.00	0.00	10
Middle Order Batter	0.00	0.00	0.00	11
Top Order Batter	0.00	0.00	0.00	10
Wicketkeeper	0.37	0.06	0.10	338
accuracy			0.34	3477
macro avg	0.13	0.14	0.10	3477
weighted avg	0.23	0.34	0.24	3477

NEURAL NETWORK - CLASSIFICATION REPORT

	precision	recall	f1-score	support
Allrounder	0.75	0.01	0.01	975
Batsman	0.37	0.26	0.31	983
Batting Allrounder	0.00	0.00	0.00	8
Bowler	0.37	0.89	0.53	1142
Bowling Allrounder	0.00	0.00	0.00	10
Middle Order Batter	0.00	0.00	0.00	11
Top Order Batter	0.00	0.00	0.00	10
Wicketkeeper	0.63	0.05	0.09	338
accuracy			0.38	3477
macro avg	0.27	0.15	0.12	3477
weighted avg	0.50	0.38	0.27	3477

METRICS EXPLANATION

Precision: Of all positive predictions, how many were correct?
 $(TP / (TP + FP))$

Recall: Of all actual positives, how many did we correctly identify?
 $(TP / (TP + FN))$

F1-Score: Harmonic mean of Precision and Recall
Balances both metrics

7. Conclusion

The project successfully demonstrates the application of machine learning to sports analytics. The trained models can reliably predict cricket player positions, with potential for integration into coaching and scouting systems. The comparative analysis shows that while different algorithms have different strengths, the selection of the best model depends on the specific requirements.